

ComicCraft - AI Comic Story Creator using Gemini Models

7. Testing and Project Results

Technology Track: Generative AI with Google

1. Testing Approach

ComicCraft was tested using a mix of functional, integration, performance, and usability checks to confirm it reliably converts story prompts into finished comics before the final demonstration.

2. Test Cases and Results

ID	Scenario	Expected Result	Outcome
T-01	Submit a normal short story idea	Comic strip generated with the correct panel count	Passed
T-02	Submit an empty story field	Validation error shown, no API call made	Passed
T-03	Change art style and regenerate	Artwork reflects the newly selected style	Passed
T-04	Regenerate a single panel	Only that panel changes; rest remain the same	Passed
T-05	Submit a very long story	System trims/summarizes input or shows a length warning	Passed
T-06	Simulate a Gemini API timeout	Clear error message with retry option shown	Passed
T-07	Download the finished comic	File downloaded matches the preview exactly	Passed
T-08	Load the app on a mobile screen	Layout adjusts responsively without breaking	Passed
T-09	Submit a prompt with unsafe content	Content safety filter blocks the request	Passed

3. Bugs Identified and Resolved

ID	Issue	Severity	Resolution
B-01	Character appearance varied slightly across	Medium	Reused one fixed

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	panels		style/character description for the whole comic
B-02	Loading indicator did not stop after an API error	Low	Added explicit error handling to stop the indicator
B-03	Layout broke on narrow mobile screens	Medium	Rebuilt using a responsive CSS grid

4. Performance Results

A standard 4-panel comic was generated in an average of roughly 40–55 seconds across repeated test runs, staying within the target of under one minute, including both text and image generation steps.

5. Usability Feedback

A small group of test users tried the application with their own story ideas. Overall feedback was positive regarding ease of use; the most common suggestion was to add more art style options, which has been logged as a future enhancement.

6. Overall Project Results

- Successfully generates coherent, illustrated multi-panel comics from open-ended text prompts.
- Maintains consistent character appearance across panels using a shared style prompt.
- Meets the target generation time for typical comics.
- Handles invalid input and API failures gracefully without crashing.

7. Summary

Testing confirmed that ComicCraft meets its functional and performance goals, reliably turning a written story idea into a finished, visually consistent comic strip, and is ready for demonstration.